

Grasp obedience across fine-tuning recipes

Standard fine-tuning

action expert + state projection

67% obeyed n = 195 grasps

Grasped orange

		Left	Middle	Right	Top right	Bottom right
Requested orange	Left	97 n=33	3			
	Middle	15	80 n=46	4		
	Right		52	41 n=71	6	1
	Top right	4			96 n=25	
	Bottom right		5	5	42	47 n=19

LM-tuned

+ language model

67% obeyed n = 206 grasps

		Left	Middle	Right	Top right	Bottom right
Requested orange	Left	92 n=24	8			
	Middle	11	89 n=63			
	Right	2	64	32 n=56	2	
	Top right	3			89 n=38	8
	Bottom right		4	8	62	25 n=24

Fully-tuned

+ language model + vision encoder

64% obeyed n = 200 grasps

		Left	Middle	Right	Top right	Bottom right
Requested orange	Left	100 n=35				
	Middle	13	78 n=55	9		
	Right	2	66	27 n=56	4	2
	Top right	6			84 n=32	9
	Bottom right	11		11	44	33 n=18

73% obeyed n = 203 grasps

Grasped orange

		Left	Middle	Right	Top right	Bottom right
Requested orange	Left	97 n=30	3			
	Middle	16	79 n=62	2	3	
	Right		48	47 n=58	3	2
	Top right				96 n=28	4
	Bottom right		5		29	67 n=21

67% obeyed n = 181 grasps

		Left	Middle	Right	Top right	Bottom right
Requested orange	Left	71 n=42	17		12	
	Middle	17	75 n=52	4	4	
	Right	10	36	46 n=39	5	3
	Top right				93 n=27	7
	Bottom right				47	53 n=17

 **Obeyed**
grasped the requested orange

 **Misgrab**
grasped a different orange

Cell value: % of the row's grasps;
darker shade = larger share.

Scene states 0-1 pooled (with two placed, one orange remains: trivial).

No fully-tuned run exists for
Teleop+Auto.

Teleop

Teleop+Auto